JAN-P-658

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NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

PACKAGING AND PACKING OF ELECTRICAL EQUIPMENT AND SPARE PARTS (ELECTRONIC, ELECTRICAL, AND ELECTRO-MECHANICAL)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–100—Packaging and Packing for Overseas Shipment—General Specification.
JAN–P–103—Packaging and Packing for Overseas Shipment—Boxes; Wood Cleated; Solid Fiberboard.
JAN–P–104—Packaging and Packing for Overseas Shipment—Crates, Sheathed, Wood, Nailed.
JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–115—Packaging and Packing for Overseas Shipment—Compound, Sealing, Dip-Coating.
JAN–P–116—Packaging and Packing for Overseas Shipment—Preservation, Methods of.
JAN–P–120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.
JAN–B–121—Barrier-Materials, Greaseproof.
JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN–P–127—Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water Resistant.
JAN–P–132—Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed.
JAN–P–133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.
JAN–B–233—Boxes, Spare Parts (Shipboard Use)

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ARMY-NAVY AERONAUTICAL SPECIFICATION

AN–C–118—Crates and Boxes, Aircraft and Airframe Component Parts (General Specification For).

U. S. ARMY SPECIFICATIONS

3–182 —Compound, Insulation, Ignition.

94–40645—Marking, Exterior, Domestic and Export Shipment, by Contractors.[1]

100–2 —Standard Specification for Marking Shipments by Contractors.[2]

ARMY AIR FORCES SPECIFICATION

40985—Marking of Interior Packages for Radio and Radar Equipment.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[3]

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

B. SCOPE

B–1. This specification provides detailed requirements for procedures to accomplish the cleaning, preservation, unit packaging, and packing of electrical equipment and spare parts (electronic, electrical, and electro-mechanical) for domestic and overseas shipment. (See par. H–1.)

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—

C–1a. *Preservatives (column 2 of table I).*—The standard types of preservatives designated by code numbers refer to specifications listed in table I of Joint Army-Navy Specification JAN–P–116. (See par. H–3.)

C–2. *Workmanship.*—All operations and processes involved in accomplishing the cleaning, preservation, unit packaging, and packing requirements specified herein, shall be in accordance with the highest grade practices associated with this type of work. All equipment used to perform the various procedures shall be such as to meet the requirements of this specification to the satisfaction of the procuring agency.

D. GENERAL REQUIREMENTS

PRESERVATION

D–1. *Preservation.*—Standard methods of cleaning and preservation shall be as described in detail in Joint Army-Navy Specification JAN–P–116 and appendixes thereto. The selection within a type of the applicable preservative referenced in Joint Army-Navy

[1] Applicable only to Air Force purchases.
[2] Applicable only to Army purchases.
[3] Applicable only to Navy purchases.

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Specification JAN–P–116 is at the option of the manufacturer unless otherwise specified in the applicable commodity or equipment specification or in the contract or order.

PACKAGING

D–2. *Packaging.*—

D–2a. *Unit packages (column 3 of table I).*—Unit packaging methods designated by code numbers shall be as specified in Joint Army-Navy Specification JAN–P–116. (See par. H–5.) Unless otherwise specified in the applicable commodity or equipment specification or in the contract or order, items shall be generally packaged one item to a package, except items used as sets (see fig. 1) or in quantities greater than one.

D–2b. *Intermediate containers.*—Unless otherwise specified, the intermediate containers shall conform to the requirements of Joint Army-Navy Specification JAN–P–108, JAN–P–120 (including table III), or JAN–P–133 (including table I). Procuring agencies may specify the number of unit packages to be packed into an intermediate container, if any. Unless otherwise specified, only like parts or sub-assemblies shall be packaged together in an intermediate container. The gross weight shall not exceed the weight limitations prescribed in the applicable container specification.

D–2b(1). *For on board spare parts.*—On board spare parts shall be furnished in spare parts boxes conforming to the requirements of Joint Army-Navy Specification JAN–B–233.

D–2c. *Arrangement of spare parts.*—

D–2c(1). Spare parts shall be arranged within containers in a compact manner. They shall be so arranged as to be held firmly in place, yet in a manner providing accessibility of any part without undue disturbance of the other parts. This does not mean to require compartments for small parts which have been bagged or cartoned.

D–2c(2). Armatures and rotors of all sizes shall be supported within the individual interior containers by means of chocks. Shafts shall be protected against direct contact with the chocks. Armatures and rotors in excess of 200 pounds, when packed in on board spare parts boxes conforming to Joint Army-Navy Specification JAN–B–233, shall be provided with a protecting case of metal as shown on figure 2.

D–2c(3). Spare parts of associated equipment may be packaged in containers with electrical spare parts, providing the spare parts thus combined are for the same application.

D–2d. *Weight.*—The net weight of spare parts (minus the box) shall not exceed 200 pounds in any one box. Where the combined weight of a set exceeds 200 pounds, such parts shall be grouped and packed in two or more boxes, numbered consecutively to show the number of boxes in a complete set of spares. An exception is made where one part weighs in excess of 200 pounds, in which case the spare part shall be individually boxed.

D–2e. *Index list of spare parts and tools.*—An index list of spare parts and tools conforming to the requirements of figure 1 shall be inserted in each container. When on board spare parts

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boxes are used, the list shall be inserted in the index list support located on the interior side of the cover of each type M spare parts box, or suitably secured to the interior side of the cover of the type W box. The index list shall be printed or typed and shall be so treated as to be resistant to oil, water, and fading. It shall give a complete itemized list of the contents of the container, the identification of the equipment to which the spare parts apply, and the name of the manufacturer. For Navy purchases, duplicate certified copies of the index list shall be furnished to the general storekeeper at the Naval Shipyard (or officer-in-charge at the shipyard), for which the material is intended. Index lists shall be attached by the cognizant Government inspector to each copy of the Material Inspection Report.

D–2f. *Marking for identification.*—Each unit package and intermediate container, in addition to any requirements of Joint Army-Navy Specification JAN–P–116, shall be clearly and legibly marked with the quantity, nomenclature of contents, stock number, and such other markings as may be necessary for identification, or as specified by the procuring agency.

D–2f(1). *Marking of spare parts.*—In addition to other markings required, spare parts shall be clearly and permanently marked, or provided with an identifying waterproof tag or label attached to the part. Stamped metal tags of corrosion-resistant material, or of metal treated to resist corrosion, may be used provided these tags are secured with corrosion-resistant wire. Small parts, or other parts which would be impracticable to mark, tag or label, may be inclosed in marked, waterproof envelopes. (These envelopes may constitute the wrapper of a unit package.) Where a small number of identical parts are packaged in a unit package as a set, only one part need be marked. Sufficient data shall be given to fix completely the identity, the use, and the equipment for which the spare parts are intended, unless this information is given on the index sheet within the container. In this case, an identification number corresponding to the index sheet number will be sufficient. Where a single number is used for identification, the identifying number shall be the Government-assigned stock number, except that when no stock number has been assigned, the identification number shall be the manufacturer's part number which completely fixes the identity of the part.

PACKING

D–3. *Packing.*—

D–3a. *For domestic shipment.*—Packing for domestic shipment shall be one of the following two methods, as specified in the contract or order:

D–3a(1). *Commercial packing.*—Unless otherwise specified, items, except on-board spare parts, preserved and packaged as described in paragraphs D–1 and D–2, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such a manner as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to point of delivery. When fiberboard shipping containers are used, they shall conform to paragraph D–3a(2)a(2).

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D–3a(2). *Service packing.*—

D–3a(2)a. *Shipping containers.*—Unless otherwise specified, items, except on board spare parts, preserved and packaged as described in paragraphs D–1 and D–2, where applicable shall be packed in one of the following types of shipping containers:

Boxes; Wood, Cleated, Solid Fiberboard—Joint Army-Navy Specification JAN–P–103.[4]
Boxes; Wood, Cleated, Plywood—Joint Army-Navy Specification JAN–P–105.[4]
Boxes; Wood, Nailed—Joint Army-Navy Specification JAN–P–106.
Boxes; Wood, Wirebound (Overseas Type)—Joint Army-Navy Specification JAN–B–107.[5 6]
Crates; Unsheathed, Nailed Wood—Joint Army-Navy Specification JAN–P–132.[6]
Crates; Sheathed, Wood, Nailed—Joint Army-Navy Specification JAN–P–104.
Boxes; Fiberboard (corrugated or solid). (See par. D–3a(2)a(2).)

D–3a(2)a(1). *Weight limitation for wood boxes.*—Except where an individual part or assembly exceeds this amount, the gross weight of wood boxes shall not exceed approximately 200 pounds.

D–3a(2)a(2). *Fiberboard boxes.*—Corrugated or solid fiberboard boxes shall be constructed as follows:

Maximum gross weight	Minimum average bursting strength	Maximum sum of inside dimensions
Pounds	*Pounds*	*Inches*
40	200	75
65	275	90
90	350	100

Bottom flaps of fiberboard boxes shall be sealed by means of a suitable adhesive or shall be metal stitched. Top flaps shall be sealed, stitched, or taped, or closed by a combination of these methods. If taped, kraft gummed tape of not less than 2½ inches width, 60 pounds basis weight shall be used.

D–3a(2)b. *Difficult loads.*—Unless otherwise specified in the applicable commodity or equipment specification or in the contract or order, difficult loads as defined in Joint Army-Navy Specification JAN–P–100, weighing up to 500 pounds shall be packed in nailed wood boxes conforming to the requirements of Joint Army-Navy Specification JAN–P–106; difficult loads of 500 pounds and over shall be packed in nailed wood boxes conforming to Joint Army-Navy Specification JAN–P–106, modified to include a skid type base and diagonal bracing, or nailed wood unsheathed crates conforming to Joint Army-Navy Specification JAN–P–132. Where unit packaging method II is used for difficult loads exceeding 500 pounds, nailed wood sheathed crates conforming to the re-

[4] Water-resistant fiberboard and plywood will not be required.
[5] Not applicable to Air Force purchases.
[6] Shall not be used for method II packages.

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quirements of Joint Army-Navy Specification JAN–P–104 shall be used.

D–3a(2)c. *Strapping.*—Shipping containers, except fiberboard boxes, shall be strapped in conformance with the requirements of the appendix of the applicable container specification. Strapping of fiberboard boxes shall conform to the requirements of the appendix to Joint Army-Navy Specification JAN–P–108.

D–3b. *For overseas shipment.*—

D–3b(1). *Shipping containers.*—Unless otherwise specified, items, except on board spare parts, preserved and packaged as described in paragraphs D–1 and D–2, where applicable, shall be packed in nailed wood boxes, nailed wood sheathed crates, or wood cleated, plywood boxes, as follows:

D–3b(1)a. Nailed wood boxes shall conform to the requirements of Joint Army-Navy Specification JAN–P–106. The gross weight shall not exceed that specified in Joint Army-Navy Specification JAN–P–100, unless the unit weight of the item exceeds this amount or unless otherwise specified in the applicable commodity or equipment specification or in the contract or order.

D–3b(1)b. Nailed wood sheathed crates shall conform to the requirements of Joint Army-Navy Specification JAN–P–104.

D–3b(1)c. Wood cleated, plywood boxes shall conform to the requirements of Joint Army-Navy Specification JAN–P–105.

D–3b(2). *Difficult loads.*—Unless otherwise specified in the applicable commodity or equipment specification or in the contract or order, difficult loads as specified in Joint Army-Navy Specification JAN–P–100, weighing up to 500 pounds shall be packed in nailed wood boxes or in wood cleated, plywood boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–106 and JAN–P–105, respectively; difficult loads of 500 pounds and over shall be packed in nailed wood boxes or in wood cleated, plywood boxes modified to include a skid-type base and diagonal bracing or in nailed wood sheathed crates conforming to the requirements of Joint Army-Navy Specifications JAN–P–106, JAN–P–105, and JAN–P–104, respectively.

D–3b(3). *Liners.*—Unless otherwise specified, shipping containers, except nailed wood crates, shall contain a sealed waterproof bag or its equivalent made from barrier-material conforming to the requirements of Joint Army-Navy Specification JAN–P–125, for case liners. The seams and closures shall be sealed with water-resistant adhesive conforming to Joint Army-Navy Specification JAN–P–140, and shall have water-resistance equal to that of the body material. Material packed in nailed wood crates shall be shrouded two-thirds of the way down with barrier-material conforming to the requirements of Joint Army-Navy Specification JAN–P–125, for case liners, and shrouds suitably secured in such a manner as to permit free circulation of air.

D–3b(4). *Strapping.*—Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification.

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D–3b(5). *Alternates.*—The following shipping containers may be used when specified in the applicable commodity or equipment specification or in the contract or order:

Boxes; Wood Cleated; Solid Fiberboard—Joint Army-Navy Specification JAN–P–103.[7]

Boxes; Wood, Wirebound (Overseas type)—Joint Army-Navy Specification JAN–B–107.[7 8]

Boxes; Fiberboard (V-Board), Exterior—Joint Army-Navy Specification JAN–P–108.

D–3c. *For domestic or overseas shipment.*—

D–3c(1). The number of unit packages or the number of intermediate containers to be overpacked in the shipping container, if any, may be specified by the procuring agency.

D–3c(2). Unless otherwise specified, on board spare parts boxes shall be overpacked in nailed wood unsheathed crates conforming to the requirements of Joint Army-Navy Specification JAN–P–132.

E. DETAIL REQUIREMENTS

E–1. Table I contains an alphabetical list of a number of representative items of electrical equipment and spare parts and general groupings thereof, designating the cleaning methods, preservative types, and unit packaging methods which shall apply. Materials and methods shown shall be used unless otherwise specified in the applicable commodity or equipment specification or in the contract or order. Table I is intended also as a guide in the selection of materials and unit packaging methods to be applied to other items of similar construction and components not listed therein.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. Inspection and tests shall be conducted in accordance with the requirements of Joint Army-Navy Specification JAN–P–116, except that the rough handling and cyclic exposure test shall be limited to method IA and method II packages regardless of the particular technique employed.

G. MARKING FOR SHIPMENT

G–1. In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Air Force, in accordance with the requirements of U. S. Army Specification 94–40645 and Army Air Forces Specification 40985; and for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—This specification covers the preservation, packaging, and packing of electronic, electrical, and electro-mechanical equipment where the electrical features and uses predominate. This specification does not apply to mechanical components of electro-mechanical equipment or to mechanical-electrical equipment where the mechanical features and uses predominate.

[7] Not applicable to Air Force purchases.
[8] Shall not be used for method II packages.

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H–2. Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Title, number, and date of the specification.
(b) Whether domestic or overseas packing is required; and if domestic, whether commercial packing or service packing is required.
(c) Intermediate and unit package quantities, if other than as described in paragraph D–2.
(d) Special marking requirements.

H–3. It is emphasized that interchangeability of preservatives applies to this specification only, and does not necessarily apply to the preservation or packaging of other classes of supply. When two or more Army services or Navy bureaus, or the Air Force specify different materials or procedures in a plant for the same purpose, those activities should jointly determine common materials and procedures where practicable. However, it is recognized that in some cases the Armed Forces requirements are of such a nature as to make certain differences necessary.

H–4. Any procuring agency may develop a supplemental tabulation of detailed preservation and packaging requirements as applicable to specific contractor or industry items, with which the contractor should conform in order to comply with the requirements of this specification. Preservation and unit packaging methods, in such tabulations, should conform to the requirements of Joint Army-Navy Specification JAN–P–116 and should not conflict with the requirements of this specification.

H–5. The unit package is the complete package which will be issued to the lowest echelon of use. Unit package requirements in this specification are intended to provide protection to the inclosed part, or parts, even though the unit package is removed from the exterior overseas shipping container and forwarded on to a lower echelon of use without repacking.

H–6. In general, table I is based upon items of less than 800 pounds gross weight. However, when unreferenced items are covered by individual service specifications, such specifications should apply.

H–7. Copies of Army Air Forces specifications may be obtained from the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton 1, Ohio. Both the title and identifying symbol number should be stipulated when requesting copies.

H–8. Copies of Army-Navy Aeronautical (AN) specifications may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton 1, Ohio (for Army and Air Force purchases); to the Commanding Officer, Naval Air Development Station, Johnsville, Pa. (for Navy purchases). Both the title and identifying symbol number should be stipulated when requesting copies.

H–9. Copies of Army-Navy Joint Packaging Instruction JPI–7 (required for Army and Air Force purchases) may be obtained upon application to the Signal Corps Procurement Agency, 2800 South 20th Street, Philadelphia 45, Pa. Both the title and identifying symbol number should be stipulated when requesting copies.

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H–10. Copies of Joint Army-Navy Specification JAN–P–75 may be obtained upon application to the Army-Navy Electronic and Electrical Standards Agency, Ft. Monmouth, N. J. Both the title and identifying symbol number should be stipulated when requesting copies.

H–11. Copies of National Military Establishment and Joint Army-Navy specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications and Standards." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Services within the Department of the Army, and the Department of the Air Force will obtain copies of National Military Establishment, Joint Army-Navy, and U. S. Army specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies.

H–12. Copies of National Military Establishment and Joint Army-Navy specifications and Joint Packaging Instructions (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and the identifying symbol number should be stipulated when requesting copies.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical).*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Adapter, cable	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Adapter, connector	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Adapter, crystal socket	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	Cushion for protection against mechanical damage.
Adapter, microphone	C–0	None	IA–8	Cushion for protection against mechanical damage.
Alternator		See appendix II		
Ammeter		See instruments		
Amplidyne		See appendix II		
Amplifier, audio		See instruments		
Anemometer indicating		do		
Arm, breaker	C–2	P–2, P–7, P–10	IA–2, IA–8	
Arm assembly, contact (brush)	C–2	P–2, P–7, P–10	IA–2, IA–8	
Armature, less than 27 cubic feet and 500 pounds net weight	C–1	None	II	
Armature, exceeding 27 cubic feet and 500 pounds net weight	C–2 (hand-wipe).	P–2, P–7, P–10	IA–1	Treat exposed ferrous metal surfaces only, except core. Exposed shafts and other unpainted exterior treated surfaces only shall be wrapped with grade A, type 1 or 2 material (JAN–B–121)[1] overwrapped with grade C, type 1 material (JAN–B–121)[1] grade C wrapper shall be brush-sealed with compound (JAN–P–115)[1], shrouded and packed in a fully sheathed ventilated crate.
Armature, motor generator		See armatures		
Armature, dynamotor		do		
Armature, relay	C–0	None	IA–2, IA–8	
Arrester, lightning, outdoor	C–0	None	O	Block within exterior container.
Arrester, lightning, indoor	C–0	None	IA–1, IA–2	Block within exterior container.
Asbestos, block	C–0	None	O	
Autosyns		See appendix II		
Ballast, fluorescent lamp		See appendix I		

[1] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Barometer		See instruments		
Barrier, arc insulating	C–0	None	O	
Base, antenna	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Base, motor	C–2	P–1	I	Grade A greaseproof wrapping not required.
Battery, dry	C–0	None	IIb	No desiccant.
Battery, storage		See JAN–P–208[2]		
Bearings, ball roller and pin		See JAN–P–197[2]		
Bearing seals	C–0	None	IA–8	
Bell, electric	C–0	None	IA–2, IA–5, IA–7, IA–8	Use method II if assemblies are not sealed and contain untreated, unpainted or unplated parts.
Belt, V-type	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Belting leather	C–0	None	IA–1, IA–2	
Bend, conduit	C–2	P–1	I	Grade A greaseproof wrapping not required.
Blades, switch, unplated	C–2	P–6	IA–8	
Blank, recording, dial recorder	C–0	None	IA–2, IA–8	
Block, capacitor	C–2	P–6	IA–2, IA–8	Clean and preserve metal surfaces only.
Block, fuse	C–2	P–6	IA–2, IA–8	
Blower, w/motor		See appendix II		
Board and strips, terminal without wiring	C–0	None	IA–2	
Body, attachment plug	C–0	None	IA–2, IA–8	
Body, choke	C–0	None	IA–2, IA–8	
Body, switch	C–0	None	IA–2, IA–8	
Book, instruction manual	C–0	None	IA–1, IA–2, IA–3, IA–8, IA–9.	
Box, pull or splice	C–2	P–1	I	Grade A greaseproof wrapping not required.
Box, starting, complete assembly	C–2	P–7	II	Oil filled compensator type shall be drained completely and drain plug replaced. Exterior surfaces of box shall be wiped clean with solvent-soaked rags and dried. Type P–7 preservative shall be applied to exterior surface by wiping or brushing.

[2] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Bracket, support	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Breaker, circtuit, complete assembly	C–0	None	II	
Breaker, circuit, aircraft	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Brush, electrical	C–0	None	IA–2, IA–8	
Brush, rigging	C–0	None	IA–2, IA–8	
Bridge, impedance, resistance		See instruments		
Bushing, conduit, metallic	C–2	P–1	I	Grade A greaseproof wrap not required.
Bushing, conduit, nonmetallic	C–0	None	O	
Button, push	C–0	None	IA–2, IA–8	
Buzzer, electrical, all types		See bell, electric		
By-pass assembly	C–0	None	II	
Cabinet, meter	C–2	P–1	I	Clean and preserve unpainted metal surfaces only. Grade A greaseproof wrapping not required.
Cam, latch, metallic	C–5[a]	P–9	IA–8	
Cams, nonmetallic	C–0	None	O	
Cams, metallic	C–5[a]	P–9	IA–1, IA–2, IA–8.	
Cap, attachment		See connector		
Cap, antenna, metal	C–1	P–1	I	Grade A greaseproof wrapping not required.
Cap, antenna, rubber	C–0	None	O	
Cap, distributor head, nonmetallic	C–0	None	IA–2, IA–8	
Cap, tube-shield	C–0	None	IA–9	
Cap, plug, grid	C–1	P–1	I	Grade A greaseproof wrapping not required.
Capacitor, fixed, all types	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Capacitor, variable		See instruments		
Carbon, arc	C–0	None	IA–9	
Case, meter		See cabinet, meter		
Case, only, tool, leather or canvas	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Cell, photoelectric		See instruments		
Charger, battery:				
Transformer type		See appendix I		
Motor-generator type		See appendix II		
Rectifyer type		See rectifyer, dry disk metal type.		
Electronic type		See instruments		

[a] C–1 and C–8 in combination may not be used as an alternate.

[JAN–P–658]

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Choke, RF, AF, audio filter	C–0	None	II	
Clamp, battery, bus-bar, cable, ground	C–2	P–6	IA–1, IA–2, IA–5, IA–7, IA–8.	
Cleat, porcelain or nonferrous	C–0	None	O	
Clip, fahnestock, fuse, tube, contact	C–2	P–6	IA–1, IA–2, IA–5, IA–7, IA–8.	
Clocks, electrical, mechanical, all types		See instruments		
Cloth, cable wrapping	C–0	None	IA–1, IA–2, IA–8.	
Coil		See choke		
Condensers, electrical		See capacitors		
Conduit, flexible, loom	C–0	None	IA–9	
Conduit, flexible, metallic	C–0	None	O	
Conduit, rigid, metallic	C–2	P–1	I	Treat and clean unpainted threads only. Grade A greaseproof wrapping not required.
Conduit, rigid, nonmetallic	C–0	None	O	
Conduits	C–2	P–1	I	Treat and clean unpainted threads only. Grade A greaseproof wrapping not required.
Connector, cable with electrical contacts	C–0	None	IA	Seal contact plugs with grade C wrapper (JAN–B–121)[4] and tape (JAN–P–127).[4] Coil and secure cable with connections inside of coil.
Connector, plug	C–1	P–1	I	For exterior ferrous surfaces only.
Contact, circuit breaker, distributor, ignition, relay switch	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Contact, set distributor	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Controller, except aircraft	C–0	P–2, P–7, P–10	II	Treat exterior ferrous metal surfaces only.
Converter, rotary		See appendix II		
Cord, switchboard, or extension	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Core, adjustable tuning	C–0	None	IA–2, IA–8.	

[4] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Counter, mechanical		See instruments		
Coupling, flexible, drive	C–2	P–7, P–10	IA–1, IA–2, IA–5, IA–7, IA–8.	
Coupling, RF		See choke		
Cover, meter, metallic	C–1	P–1	I	Grade A greaseproof wrapping not required.
Crystal units		See JAN–C–16[5]		
Cup, oil or grease	C–2	P–2, P–7	I	Grade A greaseproof wrapping not required.
Cut-out, distribution, line		See fuse		
Damper, rubber	C–0	None	O	
Dial, calibrated, fixed	C–0	None	IA–2, IA–5, IA–7.	
Dial glass	C–0	None	IA–1, IA–2, IA–8.	
Diaphragm, microphone	C–0	None	IA–2, IA–8.	
Dipole, antenna	C–0	None	IA–2, IA–8.	
Distributor, complete assembly	C–0	P–7	IA–1, IA–2, IA–5, IA–7, or II.	Treat untreated ferrous metal surfaces only. If method II is used, no preservative compound required.
Door, access	C–1	P–1	I	
Dynamotor		See appendix II		
Earpiece		See headset		
Electrode	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Electrolyte		See JAN–P–207[5]		
Element, thermostatic		See instruments		
Element, heating, coil type	C–0	None	II	
Element, immersion heater	C–0	None	O	
Equalizer, line loss	C–0	None	II	
Elevator		See mechanism		
Fan, electric		See appendix II		
Felt, friction, lubricating		See bearing seal		
Field coil, electric motors		See armatures		
Filter, AC power	C–0	None	II	
Filter, keying		See capacitor		
Fitting, conduit		See conduits		
Fork, tuning, frequency control		See instruments		
Frame, antenna, metal	C–1	P–1	I	
Frame, sw-board, without electrical connections	C–0	None	O	

[5] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Frame, sw-board, with electrical connections	C–0	None	II	
Frequency changers (rotary)		See appendix II		
Fuse, cartridges	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Galvanometer		See instruments		
Gaskets, all types	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Gauge, power supply		None	IA–2, IA–8.	
Gears, all types, precision and semiprecision	C–5[6]	P–2, P–7, P–10	IA–1, IA–2, IA–5, IA–7, IA–8.	Clean and preserve metallic surfaces only. Adequately reinforce or cushion gear teeth with at least 2 thicknesses of grade C, type 1 material (JAN–B–121).[7]
Gears, all types, nonprecision	C–2	P–1	I	
Gear box, complete assembly	C–2	P–7, P–10	IA–1, IA–2, IA–5, IA–7.	Interior to be completely drained following cleaning and flushing, and drain plugs replaced.
Generator		See appendix II		
Generator, hand operated		do		
Generator, with prime mover		See mechanism		
Glass, dial, meter, plexiglass		See dial glass		
Glove, leather		See belting, leather.		
Gloves, rubber	C–0	None	O	
Gong, electric		See bell, electric		
Governor, electric motor		See appendix II		
Grommet, rubber	C–0	None	O	
Guide, wave, noncritical	C–0	None	IA–2, IA–3, IA–8.	
Guide, wave, critical	C–0	None	II	
Header, antenna	C–0	None	II	
Heater, immersion w/controls	C–0	None	IA–2, IA–8.	Controls only.
Heater, space, blower-type		See appendix II		
Heater, strip	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8, II.	

[6] C–1 and C–8 in combination may be used as an alternate.

[7] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

[JAN–P–658]

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Handle, lever, switch panel	C–2	P–1	I	Preservative compound shall be applied to metallic surfaces only. Grade A greaseproof wrapping not required.
Hardware, kit	C–2	P–1	I	Grade A greaseproof wrapping not required.
Headset and handset, radio and telephone	C–0	None	II	Cushion within primary container with an approved cushioning material.
Harness, wiring, shielded and nonshielded	C–0	None	IA	Seal all openings with tape (JAN–P–127).[8] Block and brace within exterior container.
Hickey	C–2	P–1	I	Grade A greaseproof wrapping not required.
Hinge, butt, strap, ferrous or nonferrous		See JPI–7[9]		
Holder, contact brush	C–0	None	IA–2, IA–8	
Holder, crystal	C–0	do	IA–8	
Holder, electrode	C–2	P–2, P–7, P–10	IA–1, IA–2, IA–5, IA–7, IA–8.	Preservative compound shall be applied to metallic surfaces only.
Holder, lamp		See sockets		
Hood, antenna, metal	C–1	P–1	I	
Hood, antenna, fabric	C–0	None	IA–4, IA–9	
Horn, alarm (motor driven)		See appendix II		
Horn, alarm (vibrator)		See bell, electric		
Hose, rubber	C–0	None	O	
Housing, modulator output	C–1	P–1	I	
Housing, distributor, bakelite	C–0	None	O	
Hydrometer, glass and rubber	C–0	None	IA–1, IA–2, IA–8.	Cushioned within primary container with approved material and/or suitable interior packing to prevent moving. Treat unpainted, untreated metal surfaces only. Grade A greaseproof wrapping not required.
Index, switch plate	C–0	P–1	I	

[8] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

[9] Copies of this instruction may be obtained as indicated in paragraph H–9 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Indicators, electrical speed		See instruments		
Inductor		do		
Insulators, ceramic (molded rubber), bushing: feed thru-knob, standoff, strain, transposition	C–0	None	O	
Instruments, indicating recording and controlling	C–0	None	II	Cushion within primary container with approved cushioning material and/or suitable interior packing to prevent movement.
Inverter, rotary		See appendix II		
Iron, electrical, soldering	C–0	None	IA–1, IA–2.	
Jack, telephone, radio banana, panel, lamp	C–0	do	IA–2, IA–5, IA–7, IA–8.	
Jars, glass or molded composition	C–0	do	IA–1, IA–2, IA–8.	
Jewel, lamp	C–0	do	O	
Joint, ball	C–2	P–2, P–7, P–10	IA–1, IA–2, IA–5, IA–7, IA–8.	
Jumper, rigid and nonrigid, metallic	C–0	P–2, P–7, P–10	IA–1, IA–2, IA–5, IA–7, IA–8.	
Junction box (only)	C–2	P–1	I	Grade A greaseproof wrapping not required.
Key, telegraph	C–0	None	IA–2, IA–8	
Key, telephone	C–0	do	IA–2, IA–8	
Key, Woodruff	C–2	P–5	IA–8.	
Lacing, belt, rawhide	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Lacing, belt, metal	C–0	None	IA–2, IA–8.	
Lamps, ballast, glow, fluorescent, incandescent, pilot, indicator	C–0	None	O	
Lantern, electric w/o battery	C–0	None	IA–2	
Lens, condensing, indicating light and projection		See dial glass.		
Lever, switch, nonplated	C–0	P–6	IA–1, IA–2, IA–5, IA–7, IA–8.	
Light, indicator, assembly	C–0	None	IA–2, IA–8.	
Lighting fixture assembly	C–0	None	IA–2, IA–8.	
Loop, head band	C–0	None	IA–2, IA–8, IA–9.	

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Loudspeaker, cone type dynamic (reproducer)	C–0	None	II	Cushion within primary container with approved cushioning material. Cushion each unit package within export container.
Lug, battery and power		See clamp, battery.		
Magneto		See appendix II		
Magneto, filter assembly	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Magnets, permanent	C–2	P–6	IA–2, IA–8	Treat unpainted surfaces only. Cushion units with approved cushioning material (single face corrugated or equal) before placing within primary container to prevent shock.
Mat, rubber	C–0	None	O	
Mechanism, hoisting, training w/motors	C–2	P–6	II	Treat unpainted, untreated metal surfaces only. Clean and treat ferrous metal surfaces which will not affect electrical mechanism by application of cleaning and rust preventive compounds.
Megaphone, electric		See headset		
Megger		See instruments		
Meters, electric radio and communication		do		
Microphones		See headsets		
Monitor, radar reception		See instruments		
Motor generator		See appendix II		
Motor generator sets		do		
Motors, electric		do		
Mounting, capacitor, coil resistor, shock	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Needle, bearing		See JAN–P–197[10]		
Needle, recording and reproduction	C–0	None	IA–2, IA–9	
Oscilloscope, universal type		See instruments		
Outer race, bearing		See JAN–P–197[10]		

[10] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Panel, control	C–5[11]	P–1 and P–6	II	[12]
Panel, fuse terminal w/o wiring		See board, terminal.		
Panel, fuse, board, terminal w/wiring		See panel, control		
Pedestal, precision	C–2	P–2	IA–8	
Pedestal, non-precision	C–1	P–1	I	
Pin, cotter, dowel		See JPI–7[13]		
Pin, insulator, cross arm	C–0	None	O	
Plate: Stator, rotar, or filter	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Plug, attachment		See connector		
Plug, banana		See jack		
Plug, radio frequency	C–0	None	IA–2, IA–5, IA–7, IA–8.	
Plug, jack strip		See jack		
Pointer, meter	C–2	P–2, P–7, P–10	IA–2, IA–8.	
Post, binding	C–0	None	IA–2, IA–8.	
Potentiometers		See instruments		
Pothead	C–0	None	O	
Pullers, bearing, pulley	C–2	P–1	I	Grade A greaseproof wrapping not required.
Puller, fuse	C–0	None	O	
Pulley, metal	C–2	P–1	I	Grade A greaseproof wrapping not required.
Push button		See button, push.		
Race, inner, bearing		See JAN–P–197[14]		
Radar, receivers, transmitters		See instruments		
Radio frequency coupling		See chokes		
Radio frequency, wattmeter		See instruments		
Radio frequency units, receivers, transmitters		do		
Reactors		See appendix I		
Reactors, filter		See filter, power		

[11] C–1 and C–8 in combination may be used as an alternate.

[12] Nonshockproof instruments or meters shall be removed and separately packed in accordance with component listing contained herein. Vacuum tubes and indicator lamps shall be removed and packed separately. Metal framing shall be cleaned by wiping nonpainted surfaces with petroleum solvent and drying with clean dry rags. Type P–1 preservative shall be applied to all surfaces thus cleaned. Panel boards composed of nonferrous materials shall be lightly cleaned with clean dry rags and treated with type P–6 preservative. Instruments, switches, bus bar, and/or metallic fittings, shall be cleaned with petroleum solvent-methanol and processed with type P–6 preservative. All movable components shall be secured, blocked, and/or braced to stationary components and/or panel boards to prevent movement. Where meters and/or instruments are removed from the board for separate packaging, connecting wires and item location shall be match-marked. Instruments and/or assemblies containing oil shall be drained and drain plug and/or cocks replaced or closed. Cleaning solvents and preservative compounds shall not come in contact with material on which they would have an adverse effect.

[13] Copies of this instruction may be obtained as indicated in paragraph H–9 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

[14] Copies of this specification may be obtained as indicated in paragraph H–11 for Army and Air Force purchases, paragraph H–12 for Navy purchases.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Receptacle, electrical		See socket		
Recorder, wire, sound		See instruments		
Rectifier, crystal (in holder)	C–0	None	II	
Rectifier, dry disk, metal type	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Reel, unit wire	C–2	P–1	I	Grade A greaseproof wrapping not required.
Reflector, lighting fixture	C–0	None	IA–2	
Reflector, searchlight	C–0	None	II	
Regulator, voltage	C–0	None	II	
Relays, all types	C–0	None	II	Cushion all projections and/or sharp corners with approved cushioning material, block within primary container to prevent movement.
Reproducers, announcing		See loudspeaker		
Resistors, all types	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8, or II.	
Retainer, fuse		See clip		
Rheostats, all types, hand operated	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Rheostats, electrically operated	C–0	None	II	
Ring, collector	C–5[15]	P–6	IA–1, IA–2, IA–5, IA–7, IA–8.	
Ring, insulating	C–0	None	O	
Ring, slip, ferrous	C–1	P–2, P–7, P–12	IA	
Ring, slip, nonferrous	C–0	None	O	
Rod, antenna, metal	C–5	P–2, P–7, P–10	IA–1	
Rod, electrode, tungsten		See contact		
Rod, ground	C–0	None	O	
Rotors, squirrel cage	C–1	P–1	I	
Rotors, wound		See armatures		
Rotors, distributor	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Rodmeters	C–1	None	IA–2, IA–5, IA–7, IA–8.	
Rubber, channels, pipe fittings	C–0	None	O	
Scale, azimuth	C–0	None	II	
Seal, gasket, nonmetallic and metallic		See gaskets		

[15] C–1 and C–8 in combination may be used as an alternate.

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Shaft, steel	C–2	P–1	I	Grade A greaseproof wrapping not required.
Shaft, insulating (composition, hard, rubber, bakelite)	C–0	None	O	
Signal, switchboard	C–0	None	II	
Shunt, electrical	C–0	None	II	
Sleeving, cotton	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Sleeving, lead	C–0	None	O	
Sleeving, rubber	C–0	None	O	
Sleeves, cylinder	C–5	P–2, P–5, P–6	IA–1, IA–2, IA–5, IA–7.	
Sleeves, oilless type	C–0	None	IA	
Slot insulation and winding supplies	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Socket, crystal tube, lamp, radio	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Solenoids, all types		See relays		
Sonar drivers, amplifiers, and other sonar electronic devices		See instruments		
Sonar, projector		do		
Spark gap	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Speaker, dynamic		See loudspeakers		
Speaker, permanent magnet		do		
Spring, contact brush	C–2	P–2, P–7, P–10	IA–2, IA–8	
Springs, wire, coil, small	C–2	P–2, P–7, P–10	IA–1, IA–2, IA–3, IA–5, IA–7, IA–8.	
Springs, wire, coil, large	C–1	P–1	I	
Springs, tension	C–2	P–2, P–7, P–10	IA–2, IA–8	
Staple, insulating	C–0	None	O	
Starter, magnetic and nonmagnetic		See controller		
Stator, motor		See armatures		
Strip, terminal, w/wiring	C–5	P–1 and P–6	II	
Stroboscope		See instruments		
Stroboscope, disk	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	
Stud, terminal	C–2	P–2, P–7, P–10	IA–2, IA–5, IA–7, IA–8.	
Suppressor, electrical noise		See resistors, fixed.		
Switch, cut-out	C–0	None	IA–2, IA–8	

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Switch, mercury	C–0	None	IA–1, IA–2	Cushion units with approved cushioning material (single face corrugated or equal) before placing within primary container to prevent shock.
Switch, toggle or knife	C–0	None	IA–2, IA–8	
Switch, rotary snap	C–0	None	IA–2, IA–8	
Switchboards		See panel, control		
Synchronizer, photoflash lamp complete	C–0	None	II	Issue quantity cushioned within corrugated sleeve; place quantity of sleeve within primary container cushion within shipping container.
Synchros		See appendix II		
Tachometer		See instruments		
Tape, friction	C–0	None	IA–2, IA–5, IA–8	
Tape, rubber, insulating	C–0	None	IA–2, IA–5, IA–8	
Test set, meter		See instruments		
Tester, battery	C–0	None	O	
Thermoammeter, nonshockproof		See instruments		
Thermoammeter, shockproof		See instruments		
Thermometer, tube type	C–0	None	IA–1, IA–2, IA–5, IA–7, IA–8.	Cushion for protection against mechanical damage, with approved cushioning material.
Thermometer, dial type		See instruments		
Thermostat, complete		See instruments		
Tip, antenna	C–0	None	IA–2, IA–8	
Tip, cord	C–2	P–6	IA–2, IA–5, IA–7, IA–8.	
Transformers		See appendix I		
Transmitter, telephone		See instruments		
Trap, air, wave		do		
Trap, wave		do		
Tube, ballast electron		See lamps		
Tubes, electronic, all types		See JAN–P–75[a]		
Tuner, RF		See chokes		
Variometer		do		

[a] Copies of this specification may be obtained as indicated in paragraph H–10.

[JAN–P–658]

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TABLE I.—*Preservation of electrical equipment and spare parts (electronic, electrical, and electro-mechanical)—Continued.*

Spare part or assembly	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Vibrator, power-pack	C–0	None	II	
Voltage regulators		See instruments		
Voltmeter, shock-proof and non-shockproof		do		
Wand, tuning	C–2	P–6	IA–2, IA–8	
Wattmeter, shock-proof and non-shockproof		See instruments		
Wiring, assembly, shielded		See harness		
Weight, switch-board cord	C–1	P–1	I	
Wheel, code		See dial, indicator		

[JAN–P–658]

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Binding margin

LIST OF SPARE PARTS AND TOOLS

(Reproduced from manufacturer's drawing No.....
Army, Navy, or Air Force No.....)

Class designation and number of vessels.....
(Space to be left blank by manufacturer)

Application..... Appliance.....

Army, Navy, or Air Force contract or shipbuilder's order No.....

Manufacturer.....

Item No.	Number of sets[1]	Number per set[1]	Name of part or tool	Manufacturer's data			Army, Navy, or Air Force drawing and piece number	
				Catalog or serial ordering data	Manufacturer's drawing and piece number			
					Piece	Drawing	Piece	Drawing
			Over-all size 8 by 10½ inches		(2)		(2)	

Identification of appliance for which spare parts are intended.

When reordering always refer to.....
.....
(Insert manufacturer's identification of orders)
Number of sheets..... Sheet No.....

[1] The terms "number per set" and "number of sets" are defined as follows:
Number per set.—The quantity of a particular item necessary for the complete assembly of one unit.
Number of sets.—The multiple or fraction of the "number per set" required to indicate the quantity of the particular item furnished as a spare part.
Example: A four-pole direct-current motor requires four field coils in its complete assembly. If two field coils are furnished as spares, the "number per set" should be four and the "number of sets" should be 2/4 or 1/2.

[2] These two columns may be combined, if necessary.

FIGURE 1.—*Form for index list of spare parts and tools.*

[JAN–P–658]

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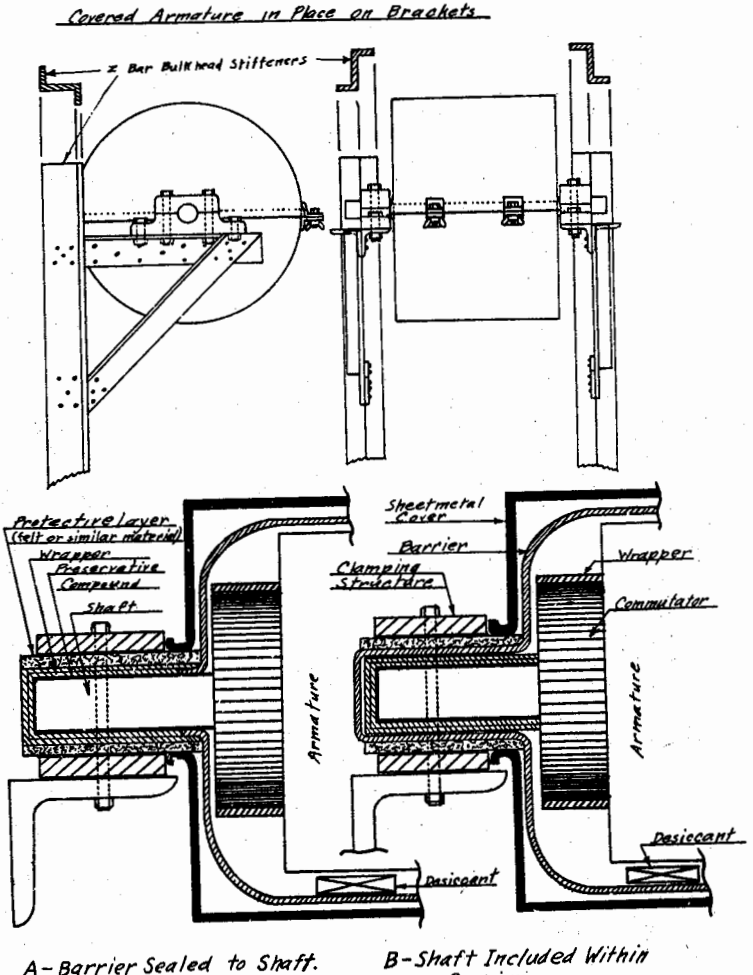


FIGURE 2.—*Mounting of armatures and rotors (in excess of 200 pounds).*

[JAN-P-658]

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APPENDIX I

PRESERVATION, PACKING, AND MARKING OF TRANSFORMERS (INCLUDING REACTORS)

1. GENERAL

1–a. Transformers (including reactors) weighing up to 1,000 pounds shall be classed as to moisture resistance, cleaned, preserved, and unit packaged as described in table I–a. Transformers (including reactors) in any class weighing more than 1,000 pounds shall be preserved and packed as specified by procuring agency.

2. APPLICABLE SPECIFICATIONS

2–a. The following specifications, of the issue in effect on date of invitation for bids, form a part of this appendix:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN–P–132—Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed.

ARMY-NAVY AERONAUTICAL SPECIFICATION

AN–C–118—Crates and Boxes, Aircraft and Airframe Component Parts (General Specification For).

3. PACKAGING

3–a. Unless otherwise specified, units weighing up to 40 pounds shall be further packaged as specified in paragraph D–2 of the specification.

4. PACKING

4–a. For domestic and overseas shipments, units packaged as specified in paragraph 3–a, shall be packed as specified in paragraph D–3 of the specification.

4–b. Units weighing over 40 pounds shall be secured by bolting within an overseas shipping container as follows:

4–b(1). Units in class I of table I–a weighing over 40 pounds, shall be packed in nailed wood unsheathed crates constructed in accordance with the requirements of Joint Army-Navy Specification JAN–P–132.

4–b(2). Class II and III units, weighing between 40 and 150 pounds, shall be packed in style 2 nailed wood boxes or wood cleated, plywood boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–106 and JAN–P–105, respectively.

4–b(3). Class II and III units weighing over 150 pounds shall be packed in intermediate crates conforming to the requirements of Army-Navy Aeronautical Specification AN–C–118.

4–c. *Ventilation.*—Intermediate crates and wood cleated plywood boxes containing transformers (including reactors) packaged method O shall be ventilated. Intermediate crates shall be ventilated in accordance with the requirements of Army-Navy Aero-

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nautical Specification AN–C–118. Cleated plywood boxes shall be ventilated with clean-cut holes ½ inch in diameter bored near the top of the side and end panels approximately 5 inches on centers.

4–d. Unless otherwise specified, shipping containers for transformers (including reactors) weighing over 40 pounds shall not be lined.

5. MARKING FOR SHIPMENT

5–a. Marking of shipping containers shall be as specified in section G of the specification.

TABLE I–a.—*Preservation of transformers.*

Description	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Class I[1]				
Pressure sealed hermetically or gasketed	C–0	P–1 on unprotected exterior ferrous metal surfaces.	O	
Class II				
Gasketed or potted: Weight up to 30 pounds	C–1	do	IA–5, IA–7, IA–8, IIb, IId.	Grade A greaseproof paper not required.
Weight over 30 pounds	C–1	do	IIa, IIb, IId	Method IIb applies to item not exceeding 40 pounds net weight.
Class III[2]				
Open core or coil or natural draft ventilated construction: Weight up to 40 pounds	C–0	do	IIb, IId	
Weight over 40 pounds	C–0	do	IIa, IId	

[1] The Army service, Navy bureau, or the Air Force may, where necessary, prescribe tests sufficient to determine pressure tightness as to the seal.

[2] Class II transformers weighing over 30 pounds and all class III transformers (including reactors) may be packaged method O if the equipment employs materials which are not adversely affected by extreme conditions of moisture insofar as corrosion and dielectric strength are concerned.

[JAN–P–658]

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APPENDIX II

PRESERVATION, PACKAGING, PACKING, AND MARKING OF ELECTRIC MOTORS, GENERATORS AND OTHER ROTATING ELECTRICAL EQUIPMENT

1. GENERAL

1–a. Electric motors, generators and other rotating electrical equipment shall be classed as to moisture resistance, cleaned, preserved, and unit packaged as described in table II–a.

2. APPLICABLE SPECIFICATIONS

2–a. The following specifications, of the issue in effect on date of invitation for bids, form a part of this appendix:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–104—Packaging and Packing for Overseas Shipment—Crates, Sheathed, Wood, Nailed.
JAN–P–105—Packaging, and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN–B–121—Barrier-Materials, Greaseproof.
JAN–P–127—Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water Resistant.
JAN–P–132—Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed.

ARMY-NAVY AERONAUTICAL SPECIFICATION

AN–C–118—Crates and Boxes, Aircraft and Airframe Component Parts (General Specifications For.)

U. S. ARMY SPECIFICATION

3–182—Compound, Insulation, Ignition.

3. PACKAGING

3–a. Unless otherwise specified, units weighing up to 40 pounds shall be further packaged as specified in paragraph D–2 of the specification.

4. PACKING

4–a. For domestic and overseas shipments, units packaged as specified in paragraph 3–a, shall be packed as specified in paragraph D–3 of the specification.

4–b. Units weighing over 40 pounds shall be secured by bolting within an overseas shipping container as follows:

4–b(1). Units in class I weighing over 40 pounds and class II units weighing over 1,000 pounds shall be packed in nailed wood unsheathed crates constructed in accordance with the requirements of Joint Army-Navy Specification JAN–P–132.

4–b(2). Class II units weighing over 40 pounds and up to 150 pounds shall be packed in style 2 nailed wood boxes or wood cleated, plywood boxes conforming to the requirements of Joint

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Army-Navy Specifications JAN–P–106 and JAN–P–105, respectively.

4–b(3). Class II units weighing between 150 pounds and 1,000 pounds shall be packed in intermediate crates conforming to the requirements of Army-Navy Aeronautical Specification AN–C–118.

4–b(4). Class II units weighing over 1,000 pounds shall be packed in nailed wood sheathed crates constructed in accordance with the requirements of Joint Army-Navy Specification JAN–P–104.

4–c. *Ventilation.*—Crates and wood cleated plywood boxes containing rotating electrical equipment packaged method O or method I shall be ventilated as follows: Intermediate crates shall be ventilated in accordance with the requirements of Army-Navy Aeronautical Specification AN–C–118. Nailed wood sheathed crates shall be ventilated in accordance with the requirements of Joint Army-Navy Specification JAN–P–104. Wood cleated, plywood boxes shall be ventilated with clean-cut holes ½ inch in diameter bored near the top of the side and end panels approximately 5 inches or centers.

4–d. Unless otherwise specified, shipping containers for units weighing over 40 pounds shall not be lined.

[JAN–P–658]

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5. MARKING FOR SHIPMENT

5-a. Marking of shipping containers shall be as specified in section G of the specification.

TABLE II-a.—*Preservation of rotating electrical equipment.*

Description	(Col. 1) Cleaning method	(Col. 2) Types of preservative	(Col. 3) Unit packaging method	Special requirements
Class I				
All degrees of submersible enclosures: All sizes	C-1	P-1 or P-2	I	See note 1.
Class II				
All degrees of enclosure except submersible: Weights up to 100 pounds	C-1	P-1 or P-2	IA, IIb	See note 2 up to 40 pounds net weight.
			IIa, IId, IIe	See note 1.
Weights 100 pounds to 1,000 pounds	C-1	P-1 or P-2	IIa	See notes 1 and 3
Weights over 1,000 pounds	C-1	P-1 or P-2	I	See note 1.

Note 1.—*Shafts and couplings.*—Shafts and couplings with unpainted surfaces shall be coated with a type P-1 preservative.

Bearings.—In general, ball or roller bearings shall be lubricated as for service. Lubricating systems, sleeve bearings, and housings shall be flushed with a type P-2 or P-10 preservative.

Commutators.—When commutators are accessible, lift the brushes and wrap a piece of grade A greaseproof barrier-material conforming to requirements of Joint Army-Navy Specification JAN-B-121 around the commutator in such a manner that it is completely covered. Tape conforming to Joint Army-Navy Specification JAN-P-127 may be used to hold the paper in place. The brushes may be permitted to rest on the grade A greaseproof barrier-material. No preservative shall be applied to the commutator or brushes. On fractional HP motors and small generators having commutators with diameter of 4 inches or less, where disassembly of the motor or generator would be necessary to comply with this requirement, commutators need not be wrapped as described above.

Slip rings.—Slip rings with diameter of more than 4 inches shall be wrapped with grade A greaseproof barrier-material conforming to Joint Army-Navy Specification JAN-B-121. Slip rings of ferrous material shall first be coated with a type P-2, P-7, or P-12 preservative. Bronze or brass slip rings require no preservative.

In general, it is preferred that the motor or generator shall not be disassembled from the unit with which it is assembled or partially disassembled. Where the size of the equipment involved dictates for the purpose of packaging or packing that disassembly is necessary, disassembly of the equipment is permissible if approved by the manufacturer. Brushes and brush rigging may, however, be disassembled and separately packaged at the option of the manufacturer.

Note 2.—An alternate packaging method may, when so specified, be applied to inclosed units in class II as follows:

Seal all openings that would permit the entrance of liquid water with tape conforming to the requirements of Joint Army-Navy Specification JAN-P-127, type I, and spray over tape and motor with compound conforming to U. S. Army Specification 8-182.

Note 3.—Class II units weighing between 100 and 1,000 pounds may be packaged method O if the equipment employs materials which are not adversely affected by extreme conditions of moisture insofar as corrosion and dielectric strength are concerned.

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